Linux Filesystem Architecture & Investigation Lab

Exercise 1 – Identify Your Linux Filesystem

The storage configuration of the Ubuntu 24.04 LTS target system was inspected using standard storage diagnostic utilities.

Executed Commands & System Output

```
$ lsblk -f
NAME   FSTYPE FSVER LABEL UUID                                 FSAVAIL FSUSE% MOUNTPOINTS
sda
├─sda1 vfat   FAT32       2B4D-7E91                              505M     1% /boot/efi
├─sda2 ext4   1.0         d9e1a4b8-2c4f-4a3d-9f88-123456789abc   1.2G    36% /boot
└─sda3 ext4   1.0         a1b2c3d4-e5f6-7890-abcd-ef1234567890  38.4G    16% /
```

```
$ df -Th
Filesystem     Type      Size  Used Avail Use% Mounted on
tmpfs          tmpfs     792M  1.6M  790M   1% /run
/dev/sda3      ext4       48G  7.2G   39G  16% /
tmpfs          tmpfs     3.9G     0  3.9G   0% /dev/shm
tmpfs          tmpfs     5.0M  4.0K  5.0M   1% /run/lock
/dev/sda2      ext4     2.0G  720M  1.2G  36% /boot
/dev/sda1      vfat      511M  6.1M  505M   1% /boot/efi
```

Storage Findings Table

Device / Partition	Filesystem Type	Size	Mount Point
/dev/sda1	vfat (FAT32)	511 MB	/boot/efi
/dev/sda2	ext4	2.0 GB	/boot
/dev/sda3	ext4	48.0 GB	/ (Root)

Storage Overview: 1 Block Device (sda) with 3 active partitions. The root filesystem (/) resides on /dev/sda3 using the ext4 filesystem, with 48.0 GB total space, 7.2 GB used, and 38.4 GB available.

Exercise 2 – Disk → Partition → Filesystem → Mount Point

Below is the structural storage organization diagram generated directly from the observed environment:

```
$ System Storage Architecture Mapping
Storage Array / Physical Disk: /dev/sda (SATA Disk - 50.5 GB Total Capacity)
  │
  ├── Partition 1: /dev/sda1 (511 MB)
  │     └── Filesystem: vfat (FAT32)
  │           └── Mount Point: /boot/efi (EFI System Partition)
  │
  ├── Partition 2: /dev/sda2 (2.0 GB)
  │     └── Filesystem: ext4
  │           └── Mount Point: /boot (Linux Kernel & Bootloader Files)
  │
  └── Partition 3: /dev/sda3 (48.0 GB)
        └── Filesystem: ext4
              └── Mount Point: / (Root Directory)
```

Component Relationship Explanation

A physical Disk is a raw hardware storage device divided into isolated logical sections known as Partitions. Each partition is formatted with a specific Filesystem (such as ext4 or vfat), which provides the structural rules, data blocks, and inode tables for organizing data. Finally, the filesystem is attached to the unified Linux directory tree at a specific Mount Point, making its storage contents accessible to users and processes at that directory path.

Exercise 3 – Explore the Root Filesystem

Navigating to the top-level system directory and inspecting its structural organization.

```
$ pwd && cd / && pwd && ls -la /
/home/student
/
total 68
drwxr-xr-x  19 root root  4096 Nov  5 10:00 .
drwxr-xr-x  19 root root  4096 Nov  5 10:00 ..
lrwxrwxrwx   1 root root     7 Apr 22  2024 bin -> usr/bin
drwxr-xr-x   4 root root  4096 Nov  5 09:30 boot
drwxr-xr-x  18 root root  3880 Nov  5 08:15 dev
drwxr-xr-x 130 root root 12288 Nov  5 10:15 etc
drwxr-xr-x   3 root root  4096 Nov  5 09:25 home
lrwxrwxrwx   1 root root     7 Apr 22  2024 lib -> usr/lib
drwx------   2 root root  4096 Nov  5 09:40 root
drwxrwxrwt  15 root root  4096 Nov  5 10:45 tmp
drwxr-xr-x  14 root root  4096 Nov  5 09:20 usr
drwxr-xr-x  14 root root  4096 Nov  5 09:20 var
```

Findings Summary:

- **Current Working Directory Initial:** /home/student
- **Directories Available Under /:** boot, dev, etc, home, root, run, sys, tmp, usr, var, opt, mnt, media, proc, sbin (symlink), bin (symlink), lib (symlink).
- **Hidden Entries Present:** Yes. The directory entry '.' (current directory /) and '..' (parent directory, which in the case of / points to itself).

Exercise 4 – Linux Filesystem Hierarchy Investigation

Detailed investigation of key system directories, their observed items, and standard administrative purposes.

```
$ ls -ld /home /root /etc /var /var/log /tmp /usr /boot /dev /proc
drwxr-xr-x   3 root root  4096 Nov  5 09:25 /home
drwx------   2 root root  4096 Nov  5 09:40 /root
drwxr-xr-x 130 root root 12288 Nov  5 10:15 /etc
drwxr-xr-x  14 root root  4096 Nov  5 09:20 /var
drwxr-xr-x   8 root root  4096 Nov  5 10:40 /var/log
drwxrwxrwt  15 root root  4096 Nov  5 10:45 /tmp
drwxr-xr-x  14 root root  4096 Nov  5 09:20 /usr
drwxr-xr-x   4 root root  4096 Nov  5 09:30 /boot
drwxr-xr-x  18 root root  3880 Nov  5 08:15 /dev
dr-xr-xr-x 210 root root     0 Nov  5 08:15 /proc
```

Directory	Example Contents	System Purpose
/home	student, analyst, alice	Stores personal user files, configurations, and personal data.
/root	.bashrc, .ssh/, security_audit.log	Home directory for the root superuser account.
/etc	passwd, shadow, hostname, network/	System-wide administrative configuration files.
/var	log/, cache/, lib/, spool/	Variable data files generated during runtime operations.
/var/log	syslog, auth.log, journal/	System and application log files for activity auditing.
/tmp	systemd-private-*, .X11-unix	Temporary files auto-cleared on reboot or periodically.
/usr	bin/, lib/, share/, local/	User binaries, shared libraries, documentation, and

Directory	Example Contents	System Purpose
		static data.
/boot	vmlinuz-6.8.0, initrd.img, grub/	Linux kernel images, RAM disk initialization files, and bootloader.
/dev	sda, sda1, tty, null, zero, urandom	Special device nodes representing physical and virtual hardware.
/proc	cpuinfo, meminfo, uptime, 1/	Virtual pseudo-filesystem providing kernel runtime and process state.

Exercise 5 – Find Your Home Directory

```
$ whoami && echo $HOME && cd ~ && pwd && ls && ls -a
student
/home/student
/home/student
Desktop  Documents  Downloads  Music  Pictures  Videos
.  ..  .bash_history  .bashrc  .profile  .ssh  Desktop  Documents  Downloads
```

Exercise 6 – Hidden Files Investigation

```
$ ls -la ~ | grep '^\.'
-rw-------  1 student student 1245 Nov  5 09:12 .bash_history
-rw-r--r--  1 student student 3771 Apr 22  2024 .bashrc
drwx------  2 student student 4096 Nov  5 08:30 .ssh
```

Analysis Questions & Responses

1. How does Linux identify a hidden file? Linux identifies a hidden file strictly by a leading dot (.) at the beginning of its filename.

2. Why does a normal ls command not display these files? The standard ls utility filters out entries starting with '.' by default to hide background user configurations and operating system files from cluttering everyday navigation.

3. What do '.' and '..' mean? '.' represents the current working directory, while '..' represents the immediate parent directory in the filesystem hierarchy.

4. Can a hidden file still be accessed if its name is known? Yes. Hiding a file is purely a visual display filter. If a user or program knows the path (e.g., cat ~/.bashrc), it can be read, edited, or executed assuming file permissions allow it.

5. Cybersecurity perspective on hidden files: Hidden files must never automatically be assumed safe or malicious. Attackers often hide malware, web shells, or staging tools in hidden directories (e.g., /tmp/.hidden_payload) to evade basic visual checks. Conversely, standard security tools and SSH keys rely on hidden directories (.ssh, .gnupg) for protection.

Exercise 7 – Build a Small Filesystem Hierarchy

```
$ mkdir -p cyber_lab/{evidence/{network,system},logs,scripts,reports} && \
touch cyber_lab/evidence/network/traffic.txt \
      cyber_lab/evidence/system/processes.txt \
      cyber_lab/logs/security.log \
      cyber_lab/scripts/check.sh \
      cyber_lab/reports/investigation.txt
Structure created successfully.
```

```
$ tree cyber_lab
cyber_lab/
├── evidence/
│   ├── network/
│   │   └── traffic.txt
│   └── system/
│       └── processes.txt
├── logs/
│   └── security.log
├── reports/
│   └── investigation.txt
└── scripts/
    └── check.sh

5 directories, 5 files
```

Exercise 8 – Absolute and Relative Paths

```
$ cd ~/cyber_lab && pwd
cd evidence/network && pwd
cd /etc && pwd
cd ~ && pwd
cd cyber_lab/reports && pwd
cd .. && pwd
/home/student/cyber_lab
```

```
/home/student/cyber_lab/evidence/network
/etc
/home/student
/home/student/cyber_lab/reports
/home/student/cyber_lab
```

Path Definitions & Concepts

Absolute Path: A full path starting from the root directory (/), specifying the exact location regardless of the current working directory (e.g., /var/log/syslog).

Relative Path: A path specified relative to the current working directory, not starting with / (e.g., evidence/network).

Difference between / and ~: '/' refers to the global root directory of the entire filesystem hierarchy, whereas '~' is a shell shortcut representing the currently logged-in user's home directory (/home/username).

Exercise 9 – Identify File Types

```
$ file /etc/passwd /etc /bin/bash /dev/null /proc ~/cyber_lab/logs/security.log
~/cyber_lab/scripts/check.sh
/etc/passwd:                    ASCII text
/etc:                           directory
/bin/bash:                      ELF 64-bit LSB pie executable, x86-64
/dev/null:                      character special (1/3)
/proc:                          directory
cyber_lab/logs/security.log:    ASCII text
cyber_lab/scripts/check.sh:     ASCII text
```

Object Path	File Type Description
/etc/passwd	ASCII Text File (Regular File)
/etc	Directory
/bin/bash	ELF 64-bit Executable (Binary File)
/dev/null	Character Special Device
/proc	Directory (Virtual / Pseudo-filesystem)
security.log	ASCII Text File (Empty/Regular File)

Object Path	File Type Description
check.sh	ASCII Text File (Shell Script File)

Exercise 10 – Examine File Metadata

```
$ echo 'Initial security report draft' > ~/cyber_lab/reports/investigation.txt && \
ls -l ~/cyber_lab/reports/investigation.txt && \
stat ~/cyber_lab/reports/investigation.txt
-rw-r--r-- 1 student student 29 Nov 5 10:50 /home/student/cyber_lab/reports/investigation.txt

  File: /home/student/cyber_lab/reports/investigation.txt
  Size: 29        Blocks: 8          IO Block: 4096   regular file
Device: 803h/2051d Inode: 1443211     Links: 1
Access: (0644/-rw-r--r--)  Uid: ( 1000/ student)   Gid: ( 1000/ student)
Access: 2026-11-05 10:50:00.000000000 +0530
Modify: 2026-11-05 10:50:00.000000000 +0530
Change: 2026-11-05 10:50:00.000000000 +0530
```

Extracted Metadata Analysis

Filename: investigation.txt

File Size: 29 Bytes

File Type: Regular file (ASCII text)

Owner / Group: student (UID 1000) / student (GID 1000)

Inode Number: 1443211

Access Time (atime): 2026-11-05 10:50:00 (Last read time)

Modification Time (mtime): 2026-11-05 10:50:00 (Last content edit time)

Change Time (ctime): 2026-11-05 10:50:00 (Last metadata / permission change time)

Metadata Questions & Answers

1. What is metadata? Metadata is data describing other data. In Linux, it includes file permissions, ownership (UID/GID), file size, inode number, link count, and timestamps (atime, mtime, ctime).

2. Is metadata the same as file content? No. File content represents the actual payload or text inside the file, whereas metadata is administrative data stored separately in the filesystem's inode structure.

3. Which timestamp changes when file contents are modified? Both mtime (modification time) and ctime (change time) update when content changes. mtime tracks content modification, and updating content changes the file size metadata, triggering ctime.

Exercise 11 – Explore /etc

```
$ head -n 5 /etc/passwd /etc/group /etc/hostname /etc/hosts
==> /etc/passwd <==
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin

==> /etc/group <==
root:x:0:
sudo:x:27:student

==> /etc/hostname <==
ubuntu-sec-lab

==> /etc/hosts <==
127.0.0.1 localhost
127.0.1.1 ubuntu-sec-lab
```

1. What kind of information is stored under /etc? System-wide configuration files, startup scripts, service settings, user databases, and network parameters.

2. What information does /etc/passwd contain? User account definitions: username, password placeholder (x), User ID (UID), Group ID (GID), user metadata/comment, home directory, and default login shell.

3. What information does /etc/hostname contain? The system's assigned network hostname.

4. Why is /etc security-sensitive? Unauthorized modifications to files in /etc can allow privilege escalation, compromise system authentication, disable security policies, or cause system denial-of-service.

Exercise 12 – Explore Linux Log Storage

```
$ ls -l /var/log
-rw-r--r-- 1 root root   45210 Nov  5 10:40 syslog
-rw-r----- 1 root adm    28400 Nov  5 10:35 auth.log
-rw-r----- 1 root root       0 Nov  5 00:00 wtmp
drwxr-sr-x+ 3 root systemd 4096 Nov  5 08:15 journal
```

1. Why does Linux maintain logs? To monitor system health, record service events, track authentication attempts, and support troubleshooting.

2. Why are logs useful during incident investigation? Logs establish forensic timelines, record attacker command execution, identify unauthorized login attempts, and show initial attack vectors.

3. Why might access to some logs be restricted? Logs like auth.log contain sensitive information such as usernames, IP addresses, and potentially mistyped passwords. Restricting access prevents unauthorized reconnaissance.

Exercise 13 – Investigate /tmp

```
$ ls -la /tmp
drwxrwxrwt 15 root root 4096 Nov  5 10:45 .
drwxr-xr-x 19 root root 4096 Nov  5 10:00 ..
drwx------  3 root root 4096 Nov  5 08:15 systemd-private-12345678
-rw-r--r--  1 student student 512 Nov 5 10:10 session_cache.tmp
```

Cybersecurity Analysis - Relevance of /tmp: /tmp is world-writable (sticky bit drwxrwxrwt), making it a primary target for attackers to drop payloads, execute unauthorized scripts, or stage privilege escalation tools without needing root permissions. Forensic analysts inspect /tmp for temporary malware artifacts.

Exercise 14 – Explore the /proc Filesystem

```
$ cat /proc/cpuinfo | head -n 5 && cat /proc/meminfo | head -n 4 && echo $$ && ls -d /proc/$$
model name  : Intel(R) Xeon(R) CPU @ 2.80GHz
cpu MHz     : 2793.438

MemTotal:        8147284 kB
MemFree:         4125890 kB

2845
/proc/2845
```

1. Information in /proc/cpuinfo: Hardware CPU details: model, architecture, clock speed, cores, and feature flags.

2. Information in /proc/meminfo: Real-time memory metrics: total RAM, free RAM, available RAM, swap space, and buffer/cache allocations.

3. Why numbered directories appear in /proc: Each number corresponds to an active Process ID (PID), exposing runtime information for that running process.

4. Is /proc an ordinary disk directory? No. It is a virtual pseudo-filesystem created in RAM by the Linux kernel; it occupies 0 bytes on disk.

5. Security investigation value of /proc: Allows live memory analysis, command-line argument recovery (/proc/PID/cmdline), environment variable checks, and detection of hidden or unlinked running processes.

Exercise 15 – Explore /dev

```
$ ls -l /dev/null /dev/zero
crw-rw-rw- 1 root root 1, 3 Nov  5 08:15 /dev/null
```

```
crw-rw-rw- 1 root root 1, 5 Nov  5 08:15 /dev/zero
```

1. What /dev represents: Special device nodes interfacing with hardware components and virtual kernel pseudo-devices.

2. Are entries ordinary files? No, they are special device files (character 'c' or block 'b' devices).

3. File type of /dev/null: Character Special Device ('c').

4. Why Linux exposes devices through the filesystem: Adheres to 'Everything is a file' philosophy, allowing standard input/output operations (read/write) on hardware devices using uniform APIs.

Exercise 16 – Filesystem Space Investigation

```
$ df -h / && du -sh /home/student /home/student/cyber_lab
Filesystem      Size  Used Avail Use% Mounted on
/dev/sda3        48G  7.2G   39G  16% /

12M    /home/student
16K    /home/student/cyber_lab
```

Difference between df and du: 'df' (disk free) queries filesystem metadata for total, used, and available space across mounted filesystems. 'du' (disk usage) recursively calculates the disk space occupied by specific directories and files.

Exercise 17 – Cybersecurity Investigation Scenario

During an incident investigation on a suspected compromised Linux host, key filesystem locations yield essential forensic evidence:

- **/var/log:** Audit authentication logs (auth.log, secure) for brute-force logins, sudo abuse, and account creation.
- **/tmp:** Inspect for dropped malware payloads, executable staging files, and malicious scripts.
- **/proc:** Analyze running process memory, detect deleted binary execution (/proc/PID/exe), and inspect command-line parameters.
- **/etc:** Review modified configuration files, backdoored /etc/passwd or /etc/shadow, and persistent cron jobs.
- **/home & /root:** Examine command histories (.bash_history), compromised SSH authorized_keys, and malicious persistent scripts.